

Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Top-Sheet.SchDoc	
File: IoT-Speaker-v1.3.PrjPcb	
Date: 3/21/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 5

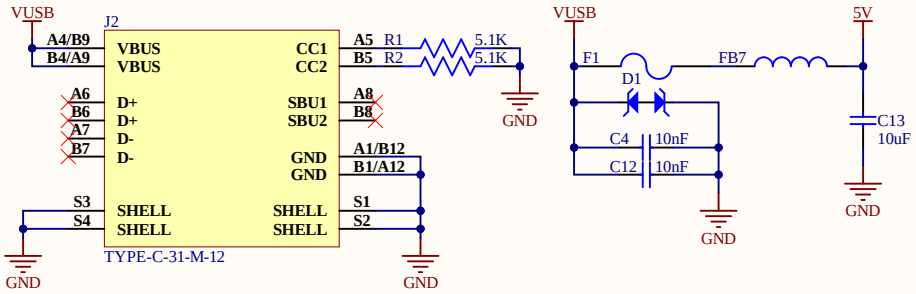
TEST POINTS



VUSB 5V CHARGE INPUT

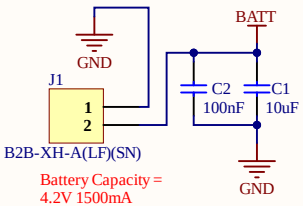
VUSB trace current 2A

5V trace current 2A



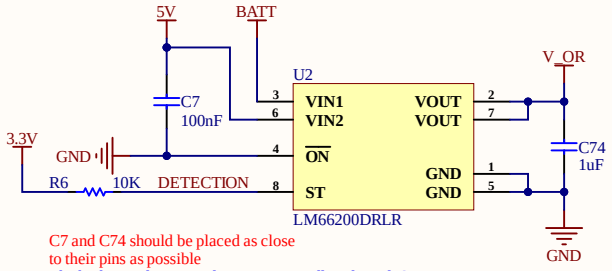
Decoupling capacitors should be placed close to each VBUS pin
ESD Diode is to be placed in manner that it has minimum impedance path with VBUS

BATTERY INPUT



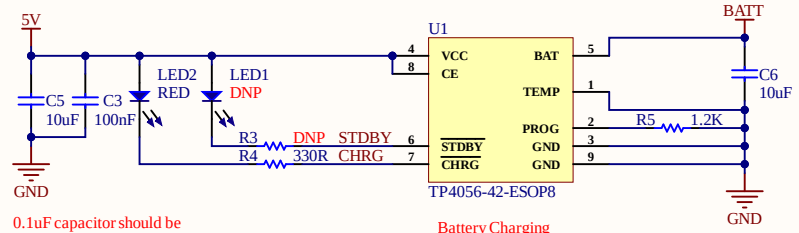
Battery Capacity = 4.2V 1500mA

POWER SUPPLY ORING CIRCUITRY



C7 and C74 should be placed as close to their pins as possible
The highest voltage supply is automatically selected. ST pin is used to detect which voltage is currently selected

BATTERY CHARGING CIRCUIT



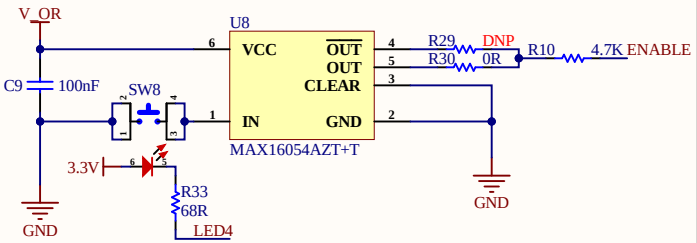
0.1uF capacitor should be placed closest to VCC pin
Place LED 2 close to USB port for visibility

Battery Charging current = 1000mA

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

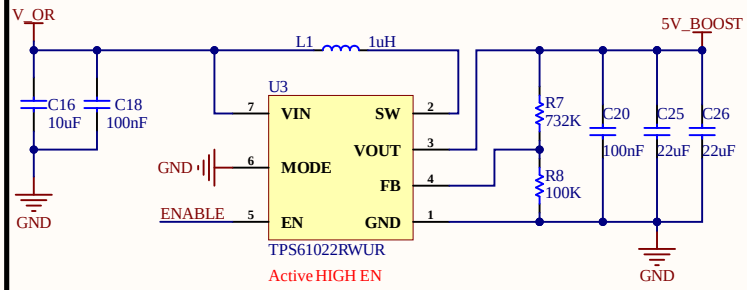
Title: Power-Sheet-1.SchDoc	
File: IoT-Speaker-v1.3.PrjPcb	
Date: 3/21/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 5

LATCH ON/OFF CONTROLLER

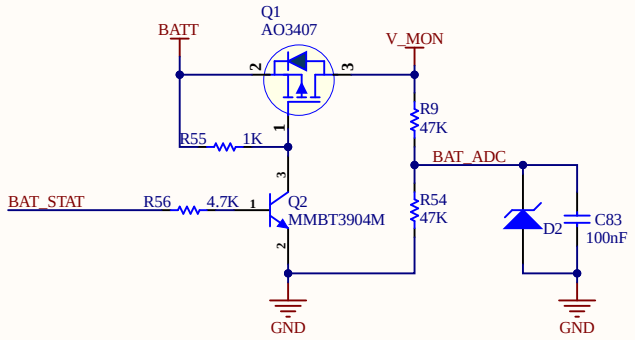


The OUT pin changes state whenever
LOW logic is received at IN pin

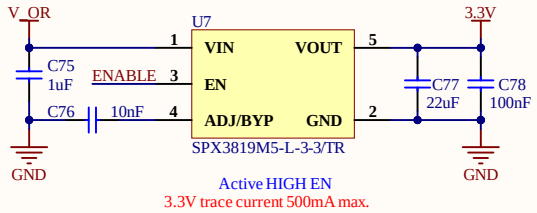
BOOST CONVERTER



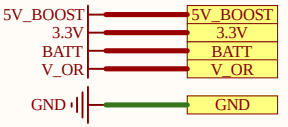
READ BATTERY VOLTAGE



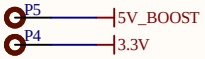
3V3 LDO



Active HIGH EN
3.3V trace current 500mA max.

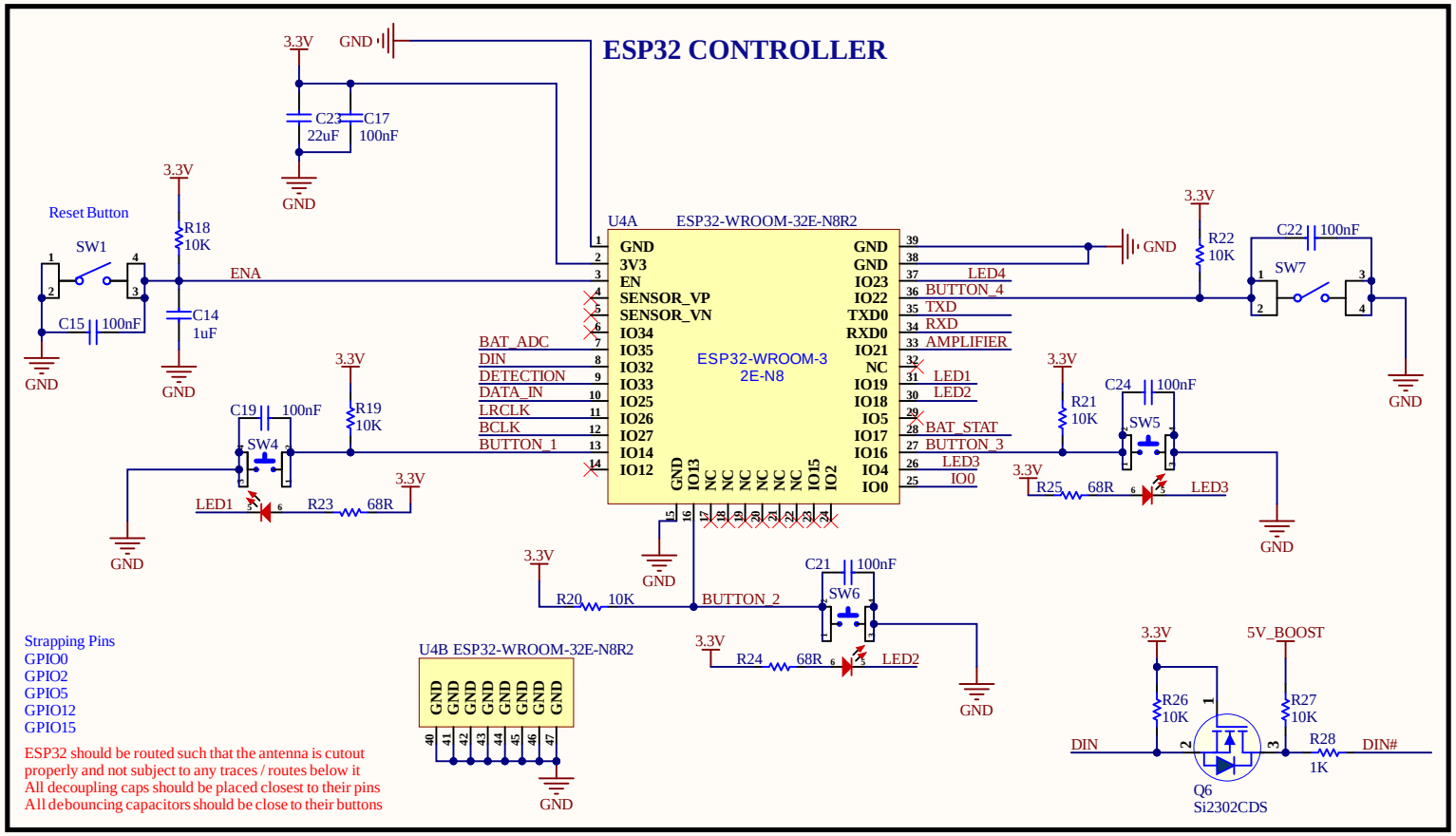


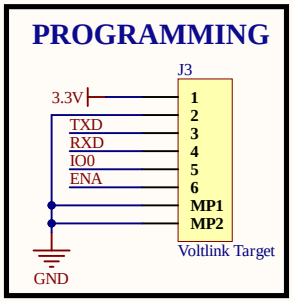
TEST POINTS

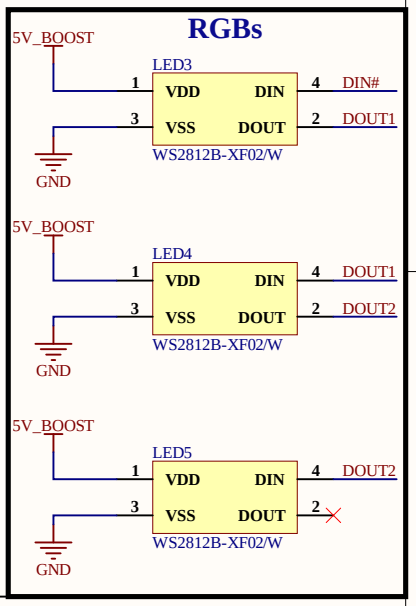


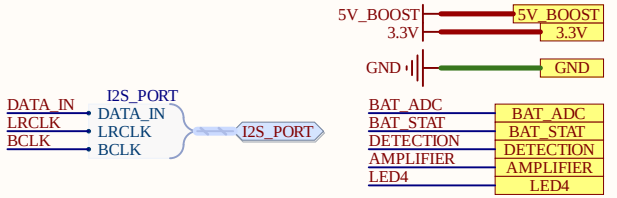
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Power-Sheet-2.SchDoc	
File: IoT-Speaker-v1.3.PrjPcb	
Date: 3/21/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 5



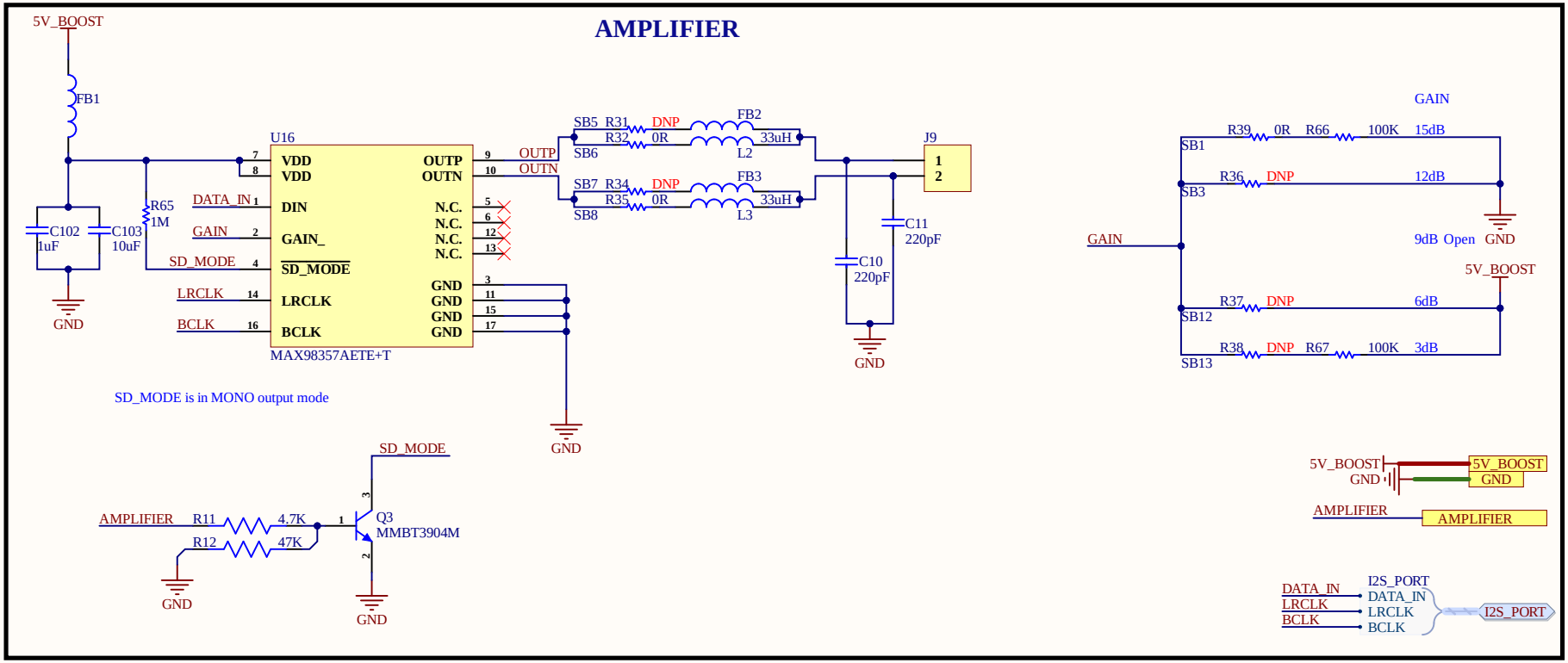






Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium
Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Controller-Sheet.SchDoc	
File: IoT-Speaker-v1.3.PrjPcb	
Date: 3/21/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 5



Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Audio-Amplifier.SchDoc	
File: IoT-Speaker-v1.3.PrjPcb	
Date: 3/21/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 5